

MOVIMIENTO Y FUNCIONES

M.C.U

MOVIMIENTO CIRCULAR UNIFORME

M.C.U → Ejemplo

LEYES CLAVE

Velocidad angular constante

$$\omega = \frac{\Delta\theta}{\Delta t}$$

Velocidad lineal

$$v = \omega \cdot r$$

$$\text{Periodo} = T = \frac{2\pi}{\omega}$$

3. Relaciones Matemáticas

$$v = \omega \cdot r \quad a_c = \omega^2 \cdot r$$

$$T = 2\pi / \omega$$

$$f = 1/T$$

4. Solución

$$v = \omega \cdot r = 5 \times 0.8 = 4$$

$$T = \frac{2\pi}{\omega}$$

$$\omega = \frac{2\pi}{T} = 1.2565$$

L.D.I

$$\text{Radio } r = 0.8 \text{ m}$$

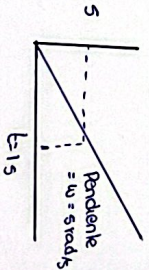
$$\text{Vel angular} = \omega = 5 \text{ rad/s}$$

$$v = ?$$

$$f = ?$$

$$T = ?$$

2. Gráfica.



FUNCIONES TRIGONOMETRICAS

Las 6 funciones

- * Sen θ
- * Cos θ
- * Tan θ
- * Csc θ
- * Sec θ
- * Cot θ

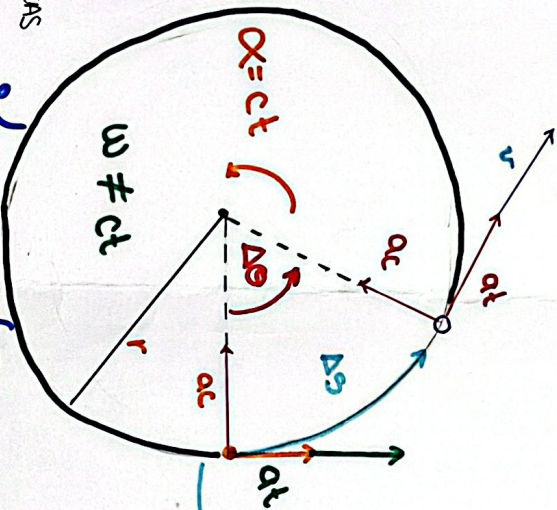
$$F = 1/T$$

$$= \frac{1}{1.256} = 0.795 \text{ Hz}$$

$$a_c = \omega^2 \cdot r = 25 \times 0.8$$

$$= 20 \text{ m/s}^2$$

Matem - Torres Sto 4B



M.C.U.A

LEYES CLAVES

$$1. \omega = \omega_0 + \alpha t$$

$$2. \theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$3. \omega^2 = \omega_0^2 + 2\alpha\theta$$

$$4. \theta = (\omega_0 + \omega)/2 \cdot t$$

Tipos de aceleración que aparecen:

TIPO

ACELERACIÓN

Angular

$$\alpha = \frac{\Delta\omega}{\Delta t}$$

Centrípeta

$$a_c = \omega^2 \cdot r$$

Tangencial

$$a_t = \alpha r$$

Cambio en rapidez de giro

Cambio de dirección

Cambio en velocidad lineal

Transformaciones de Unidades

Angulares

Ejemplo M.C.U.A

Conversiones clave

$$360^\circ = 2\pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad} \approx 57.3^\circ$$

$$1 \text{ rev} = 360^\circ$$

$$= 2\pi \text{ rad}$$

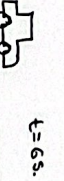
Vel

angular

$$\omega = \frac{\Delta\theta}{\Delta t}$$

$$t = 6 \text{ s}$$

Gráfico



Relaciones Matemáticas

$$\omega = \omega_0 + \alpha \cdot t$$

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$a_t = \alpha \cdot r$$

Solución

$$\omega = 0.4 \times 6 = 2.4 \text{ rad/s}$$

$$\theta = 0.6 + \frac{1}{2} \times 1 \times 36$$

$$= 22.2 \text{ rad}$$

$$f = 2\pi / (2\pi) = 1.146 \text{ rev}$$